MIL-T-8231A (USAF)
2 September 1970

SUPERSEDING
MIL-T-8231(USAF)
24 March 1953

MILITARY SPECIFICATION

TUBES: COPPER-SILICON-BRONZE, SEAMLESS, FOR AIRCRAFT

1. SCOPE

1.1 Scope. This specification covers one composition of soft annealed seamless copper-silicone bronze tubes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-C-3993	Copper and Copper-Base Alloy Mill Products; Packaging Of

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods
FED-STD-185	Continuous Identification Marking of Copper and Copper-Base Alloys

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of the specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4710

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

E-8 Tension Testing of Metallic Materials.
E-243 Rec. Practice for Electromagnetic (Eddy Current) Testing of Seamless Copper and Copper-Alloy Heat Exchanger and Condenser Tubes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The material used shall be of such quality as to produce tubing conforming to all the requirements of this specification.

3.1.1 Chemical composition. The chemical composition shall be as specified in table I.

Table I. Chemical composition 1/

Elements	Percent
Silicon	2.8 - 3.5
Manganese or Zinc	1.5 (max.)
Iron	0.24 (max.)
Lead	0.05 (max.)
Other Elements	0.5 (max.)
Copper	Remainder

1/ Analysis shall be regularly made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that these other elements are not present in excess of the above limits.

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3.2 Tensile strength. The tensile strength shall be as follows:

Tensile strength pounds per square inch, (Min)	50,000
Elongation, percent in 2 inches (min)	35

3.3 Dimensions and permissible variations

3.3.1 Diameter. The tubes shall be of the section specified, uniform in dimension throughout, and within the tolerances specified in table II.

Table II. Tolerances

Nominal outside diameter, inch	Permissible variations, (inch-plus or minus)
up to 5/8 Inclusive	.0025
Over 5/8 to 1 Inclusive	.003
Over 1 to 2 Inclusive	.004
Over 2 to 3 Inclusive	.005
Over 3 to 4 Inclusive	.006
Over 4 to 5 Inclusive	.008
Over 5 to 6 Inclusive	.009
Over 6 to 8 Inclusive	.010
Over 8 to 10 Inclusive	.013

3.3.2 Wall thickness. The tolerances given in table III will be permitted on wall thickness.

Table III. Tolerances - (inch - plus or minus).

Nominal wall thickness, inch	Diameter inch						
	1/32 to 1/8	Over 1/8 to 5/8	Over 5/8 to 1	Over 1 to 2	Over 2 to 4	Over 4 to 7	Over 7 to 10
Under - .018	.0025	.0015	.002	.0025	-	-	-
.018 - .024	.004	.0025	.0025	.003	-	-	-
.025 - .034	.004	.003	.003	.004	.005	-	-
.035 - .057	.004	.004	.0045	.0045	.0065	.009	-
.058 - .082	-	.0045	.005	.005	.0075	.010	.013
.083 - .119	-	.005	.0065	.0065	.009	.011	.014
.120 - .164	-	.007	.007	.0075	.010	.013	.015
.165 - .219	-	-	.009	.010	.013	.015	.018
.220 - .283	-	-	.012	.013	.015	.018	.020
.284 - .379	-	-	-	.015	.018	.020	.023
.380 - and over percent	-	-	-	6	6	8	8

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3.3.3 Length. Tubing of all sizes may be ordered to exact lengths or in lengths expressed as a multiple of a definite unit, with the tolerances shown in table IV.

Table IV. Tolerances (inch - plus only).

Specified lengths (feet)	Diameter inch			
	Up to 1 Incl	Over 1 to 2 Incl	Over 2 to 4 Incl	Over 4
Straight Lengths				
Up to 1/2 incl	1/32	1/16	1/16	-
Over 1/2 to 2 Incl	1/16	3/32	3/32	1/8
Over 2 to 6 Incl	3/32	1/8	1/8	1/4
Over 6 to 14 Incl	1/4	1/4	1/4	1/4
Over 14	1/2	1/2	1/2	1/2
Coils				
Up to 50 Incl	12	12		
Over 50	24	24		

3.3.3.1 When exact or multiple lengths are not ordered, tubes will be accepted in mill lengths of 7 to 14 feet, but not more than 20 percent of lengths shorter than 9 feet shall be furnished.

3.4 Flattening properties. Tubing shall be capable of being flattened flat under a gradually applied load without cracking.

3.5 Hydrostatic Pressure Resistance. Tubing shall not bulge, leak, crack, or reveal other defects when subjected to an internal hydrostatic pressure calculated according to the formula:

$$P = \frac{1.9ST}{D - T}$$

Where: P = test pressure in pounds per square inch,
T = minimum wall thickness (nominal wall thickness minus maximum negative tolerance) in inches,
S = minimum yield strength of 17,000 psi
D - nominal outside diameter of the tube, in inches.

Note (1) A diametric permanent set not to exceed 0.001 inch. per inch of tubing diameter is acceptable.

Note (2) If calculated value of P is greater than 3000 psi, actual test pressure need not be over 3000 psi.

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3.6 Flaring. Tubing shall be capable of being flared to a permanent expanded OD not less than 1.2 times the original nominal OD, without formation of cracks or other visible defects.

3.7 Identification of product. Item identification marking shall be in accordance with FED-STD-185 (see 6.3).

3.8 Tolerances. The tolerances of tubing in sizes falling outside the limits shown in this specification shall be as specified in the contract or purchase order.

3.9 Workmanship. The tubing shall be seamless, uniform in quality and temper. The exterior and interior surfaces shall be clean, smooth and free from seams, slivers, laminations, folds, grooves, cracks, and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to performs any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified, sampling, inspection, and testing shall be in accordance with Fed. Test Method Std. No. 151. All tests required herein are classified as acceptance tests and shall be performed on individual lots which have been submitted under a contract or order.

4.2 Sampling

4.2.1 Inspection lot. An inspection lot shall consist of tubing of the same size and wall thickness submitted for inspection at one time.

4.2.2 Chemical samples

4.2.2.1 Samples for check chemical analysis shall be selected as described in method 111 or 112 of Fed. Test Method Std. No. 151 to represent each 5,000 pounds or less of metal comprising the inspection lot.

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4.2.2.2 When chemical composition is determined by the spectrographic method the sample for spectrographic analysis shall be taken in accordance with method 112 of Fed. Test Method Std. No. 151.

4.2.2.3 Samples for check chemical analysis of individual inspection lots may be waived at the discretion of the inspector, provided that the manufacturer's method of composition control is acceptable to the inspector, or that the composition of the material under inspection has been determined and found to be in conformance with the chemical composition specified herein.

4.2.3 Tensile test and flattening test samples. For material having a nominal weight of less than 3 pounds per linear foot, one tensile test sample and one flattening test sample shall be selected from each inspection lot of 500 pounds or less; for inspection lots greater than 500 pounds, one additional sample shall be taken from each 1,000 pounds or fraction thereof in excess of the first 500 pounds. For material having a nominal weight of 3 pounds or more per linear foot, one tensile test sample and one flattening test sample shall be selected from each lot of 500 feet or less and one additional sample shall be selected from each 1,000 feet or fraction thereof in excess of the first 500 feet. Only one tensile test specimen and one flattening test specimen shall be taken from a given piece when more than one piece is available.

4.2.4 Hydrostatic pressure test samples. For the purpose of hydrostatic testing only, a number of lengths of tubing equivalent to 0.2 percent, or closest percentage thereto of the shipment shall be selected at random for the test prescribed in 3.5.

4.2.5 Flaring test samples. Five samples from each lot shall be taken for the flaring test.

4.3 Preparation of specimens

4.3.1 Tensile test specimens. Tensile test specimens shall be prepared as described in ASTM E8.

4.3.2 Flattening test. The ends shall be machined normal to the longitudinal axis of the specimen.

4.3.3 Flaring test. The end of the specimen to be flared shall cut square, with the cut end smooth and free from burrs, but not rounded.

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4.4 Tests

4.4.1 Chemical analysis. Chemical analysis shall be by standard wet chemical or spectrographic methods. In case of dispute, chemical analysis by standard wet chemical methods shall be the basis for acceptance.

4.4.2 Tensile test. Test shall be conducted in accordance with ASTM E8.

4.4.3 Flattening test. A test specimen consisting of a full section of tubing not less than 2 inches in length, shall be flattened between parallel plates by a gradually applied load.

4.4.4 Hydrostatic pressure test. Tubing when subjected to the hydrostatic pressure calculated by the formula in 3.5 shall be free from leaks and bulges.

4.4.5 Flaring test. The specimen shall, at room temperature, be forced axially with steady pressure over a hardened and polished tapered steel pin having a 74 degree included angle to produce a flare. The specimen and pin shall be clean and dry during test.

4.4.6 Alternate Eddy current test. In lieu of the hydrostatic test (see 4.4.4). Tubing shall be 100 percent tested at the mill by nondestructive electrical test in accordance with ASTM E243.

4.5 Reheat treatment. When failure of any lot of material to meet the requirements of this specification is due to inadequate heat treatment, the material may be reheat-treated and resubmitted for test. Only two such reheat treatments shall be allowed.

4.6 Rejection and retest. Failure of a specimen to meet specified requirements shall cause rejection of the materials represented. At the discretion of the contractor/supplier retest will be permitted. A retest sample of five specimens, one from each of five tubes, shall be tested to replace each failed specimen from the original sample. If one of the retest specimens fails, the material represented shall be rejected and no further retesting permitted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Unless otherwise specified preservation and packaging shall be level A, B or C in accordance with the requirements of MIL-C-3993.

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5.2 Packing: Unless otherwise specified packing shall be level A, B or C in accordance with the requirements of MIL-C-3993.

5.3 Marking of shipments. Interior containers and shipping containers shall be marked in accordance with MIL-STD-129. In addition, to any special marking required by the contract or order, include the following information:

COPPER-SILICON-BRONZE, SEAMLESS, FOR AIRCRAFT
MIL-T-8231
Size and thickness
Heat No.
Quantity contained (as defined in the contract or order for each size and thickness)
Name of Manufacturer or Name of Contractor (if different from manufacturer)
Contract or order No.

6. NOTES

6.1 Intended use. Seamless Copper-Silicon Bronze Tubes are intended for use on aircraft for fuel, oil, water and air lines.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification

b. Outside diameter and wall thickness.

c. Length (exact or multiple length desired).

d. Whether material is to be packaged or packed by levels A, B, or C (see 5.1 and 5.2).

e. Maximum gross weight of container.

6.3 The requirements for item identification marking (see 3.7), for packaging (see 5.1), packing (see 5.2), and marking for shipment (see 5.3) specified herein apply to direct shipment for Government activities, and apply also, when specified, to contracts or orders between the manufacturer and the Government prime contractor.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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